SONAL AMARASEKERA

+94760277717 | amarasekera.sonal@gmail.com
www.linkedin.com/in/sonalamarasekera
www.github.com/sonalamarasekera



SUMMARY

Artificial Intelligence and Machine Learning Engineer recently graduated with a Master's degree in Information and Communication Engineering having proven hands-on experience designing and maintaining end-to-end ML pipelines across full development lifecycle from data preparation, feature engineering and model training to containerized deployment and performance monitoring. Proficient in Python with limited working knowledge of SQL and C++. Comfortable collaborating with cross-functional engineering teams and communicating technical findings to varied audiences.

EDUCATION

MSc in Information and Communication Engineering (Graduated Jan 2026)
Beihang University | Beijing, China
Thesis: Latent Domain Speech Separation using Neural Audio Codec tokenization and RWKV Attention

BSc in Communication Engineering (Graduated June 2021)
Beijing Jiaotong University | Beijing, China
Thesis: Environment Monitoring and Control System using Arduino

EXPERIENCE

Graduate Machine Learning Researcher
Beihang University | Beijing, China | Sep 2023 – Dec 2025

- Designed and implemented a deep learning model using RWKV architecture and linear attention that separates mixed speech signals in the latent domain using neural audio codec tokenization. This was the final graduation project.
- Implemented various cutting-edge Machine Learning, Deep Learning and Generative AI models introduced in research papers. Contributed to group projects that involved Large Language Models, Computer Vision Models and Clustering Models.

Associate Consultant (L2)
Rizing Lanka Pvt Ltd | Colombo, Sri Lanka | Nov 2022 – May 2023

- Maintained SAP ERP systems and supporting databases across local and cloud environments across major companies in Sri Lanka. Performed structured error-log analysis to reduce routine patching incidents.

Application Support Engineer
Duo Software Pvt Ltd | Colombo, Sri Lanka | Sep 2021 – Sep 2022

- Contributed to the deployment and maintenance of a call-center system at scale across major companies in Sri Lanka using containerization and version controlling. Diagnosed VOIP instability using Wireshark packet analysis.
- Supported administration of PostgreSQL and MongoDB databases on Ubuntu/Debian production servers. Maintained Redis pipelines for real-time queuing. Load balancing for multi-scale call handling.

PROJECTS

Latent Domain Speech Separation using Neural Audio Codec Tokenization and RWKV Attention
(MSc Thesis)

- Designed and implemented the pipeline to train an RWKV based speech separation model in PyTorch. Pipeline includes data preparation, tokenization, feature engineering, model training and parameter adjusting, data shape/type handling, generating separated speech, evaluation and reproducibility. Surpassed the SOTA model at the time by 32% (Perceptual).
- Tools used: PyTorch, RNNs, RWKV, Neural Audio Codec Modelling, GenAI, Git, Pandas, NumPy, Docker, Linux, Python, Bash, TensorBoard.
- Code at: https://github.com/sonalamarasekera/quant-demix-rwkv

Sequential Voice Agent *(Personal Project)*

- Implemented a pipeline using open-source Speech-To-Text (ONNX) and Text-To-Speech (Vits) models to automatically recognize the instructions provided by a user, understand it and reply using an LLM and then give a speech output. This pipeline is made to be capable of running on edge devices. It is still a work in progress.
- Tools used: Python, ONNX, Vits, ASR, STT, TTS, Git, Linux, FastAPI
- Code at: https://github.com/sonalamarasekera/speech-pipe

BERT Fine-tuning for NLP Tasks *(Coursera Guided Projects)*

- Fine-tuned BERT for sentiment analysis using a PyTorch implementation, applying deep learning techniques to a text classification task.
- Fine-tuned BERT for text classification using TensorFlow and Keras, loading pre-trained model weights via Hugging Face Transformers.
- Tools used: PyTorch, TensorFlow, Keras, BERT, Transformer Architecture, Hugging Face

Dynamic Multi-LLM RAG Knowledge Base *(Personal Project)*

- Designed a full-stack RAG Knowledge base for a HR Policy Documentation with a stateless backend using FastAPI and LangChain. Per-request routing enabled via an LLM provider (Google, Anthropic, OpenAI) of choice. Utilizes local HuggingFace embeddings. ChromaDB vector store configured for storage and semantic understanding of the policies.
- Implemented strict hallucination guardrails (rejects out-of-scope queries with 100% reliability) by means of advanced prompt engineering and LCEL pipelines. Also built a simple decoupled Streamlit frontend that is non-tech user friendly. It is still a work in progress.
- Tools used: Python, FastAPI, Streamlit, LangChain, ChromaDB, HuggingFace, LLM, Prompt Engineering
- Code at: https://github.com/sonalamarasekera/dynamic-rag-knowledge-base

CERTIFICATIONS

- Machine Learning Operations (MLOps) on Google Cloud Specialization (2026)
- AI Foundation — Innovation Talents Science Training Program (2024)
- Fundamentals of Red Hat Enterprise Linux (2021)

LANGUAGES

- English (Fluent)
- Sinhala (Native)
- Mandarin (Working Proficiency)